

$$12.) x = a + b$$

$$y = b + a$$

$$z = x * 1$$

$$p = y * 1$$

$$q = z + p$$

Apply Algebraic Laws:-

- Commutative Law:-

$$a + b = b + a$$

$$\text{So, } x = a + b$$

$$y = b + a \text{ // same as } x$$

x and y are equivalent

- Identity Law:-

$$x * 1 = x$$

$$\text{So, } z = x$$

$$p = y$$

Assign value numbers:-

Expression

Value

a

1

b

2

$a + b$

3

$$\text{So, } x = 3$$

$$y = 3$$

$$z = 3$$

$$p = 3$$

Simplify final expression:-

$$q = z + p$$

Since both are same:-

$$q = t + t \text{ (where } t = a + b)$$